

*The Language of Business Analytics Research & Terminology

Foundational Concepts of Analytics

Business Analytics

- It is the application of data to make informed business decisions.
- Business analytics helps companies to collect data, analyze it and make recommendations based on the data collected.
- It support better and faster decision making
- BA helps organisations to make informed business decisions and improves performance.

Example

BA works on a capital APP feature like send money while adding a beneficiary, they work with compliance and the core banking system.

Question 2 Data Analytics

Data Analytics is a process where data is examined to help find trends and make informed decisions.

→ Data is collected, transformed and organized to make decisions or conclusions.

- It summarizes the historical data.

- Data Analytics - cleans, transforms raw data find patterns and make decision.

Example

Bortumelo a capital customer usually spends R1000-00 at Spar in Gauteng, Midrand.

→ Capital collects every transaction

→ The Analyst checks the pattern

→ At 02:00am a transaction of R10000 with ~~add~~ withdrawal in Limpopo and R3000 at a liquor store.

→ System flags it as 95% likely fraud and instantly blocks the card.

Source Tech Target

Question 3. Business Intelligence.

Business Intelligence is a technology operated processes for analyzing data and presenting actionable information to help managers make better decisions.

- BI - pulls data from distant source, analyze it and present
- Making informed decision based on ~~data~~ evidence.
- Uses raw data and turn into a story that users that easy understand.

Example

Capital Bank - How to stop fraud

- BI analyzes transaction patterns in real time
- card is used in midland at 13:00 and in Limpopo at 13:40
- Traces why customers close accounts
- Decision - Block suspicious transaction.

Reference

- Longvist A and Pittman V (2006)

The measurement of business intelligence.

- Domo (2024) 17 real world example where Business Intelligence makes a difference.

Question 4. DATA-DRIVEN Decision Making

DDDM is the process of making decisions based on data.

- Making decision based on evidence provided not just feeling.

- DDDM is when you use the report to change your decision.

Example: Capital analyzes customer transaction data and notice that customers who make 4 plus ATM withdrawals in a week are more than 40% are are more likely to close the account so it makes an offer to those customers not to close the account.

Question 5. BIG DATA

Big Data refers to huge datasets that cannot be handled by traditional data management systems.

- Big Data is generated from social media platforms like Twitter, Instagram and online transactions

- Velocity, Variety, Volume, Value and Veracity are the key characteristics of Big Data.

Example: when you click on your UberEats app, TikTok and Instagram DMs and the till slips in a year.

Question 6. Descriptive Analytics what happened?

DA - Summarizes past data to understand what happened.

DA does not explain what will happen next or why.

- It uses data visualization to identify relationship in datasets.

Example: A Bank uses descriptive analysis to summarize historical data.

Capital Bank uses monthly report summarizing total withdrawals in the last 3 months to see what happened.

Question 7 Diagnostic Analytics - why did it happen?

Diagnostic Analytics - investigates what actually happened within historical dataset.

- It uncovers the root causes and relationship with historical data.

- Diagnostic Analytics uses Data mining to investigate why the particular outcome has occurred.

Example Capital One saw withdrawals increased in August. Using Diagnostic Analytics they found out that monthly bank charges went up (increased) by ₹70-00 so customers withdrew their money to avoid fees.

Question 8 Predictive Analytics what is likely to happen?

- Uses historical data, machine learning techniques and statistical algorithms to estimate future outcomes

- Organizations make informed decision by checking trend of the past data.

Example Based on the last 3 months withdrawals if bank charges stay high, it predicts that Capital One more customers in the next 2 months, e.g. it shows a 10% increase in withdrawals each time charges went up.

Question 9 Prescriptive Analytics what to do?

- when data is analyzed to recommend solution in order to achieve best outcomes.

- Uses predictive analysis and recommends what to do.

Example Capital One should lower bank charges from ₹70 to ₹50-00 and offer 5% interest on balance. If data shows would reduce withdrawals by 15% and keep customers.

Question 10. DATA LITERACY

- Data literacy is the ability to read, understand, analyze and communicate with data to make better decision.

- Being able to interpret data in order to make informed decision.

example when a manager looks at the pattern of withdrawals going up understands why so much withdrawal and explain it to the team without using confusing words.

• Data measurement & Reporting.

Question 11 STRUCTURED DATA

Structured Data refers to data stored in a table with rows and columns.

- Has a fixed format

- Easy to Search, filter, analyze.

example capita's customer table, Name, account no what it is not - voice notes, not photos

Question 12 UNSTRUCTURED DATA

- Data that have no fixed table

- Doesn't fit in rows

- Need special tools to analyze

- No consistent format.

example whatsapp, Messages, emails, Instagram comments about capita

what is not - Not in rows and columns.

Question 13. DATA WAREHOUSE

Data warehouse is a central store for all clean historical data of a company.

- It is built for reporting, houses multiple databases.

Example: capita's data warehouse combines data from

ATMs, app, branches and call centre

, It is not daily operational system where transactions happen live.

Question 14. DATA MINING

Data Mining means a process of digging through huge amounts of data to find hidden patterns that

People won't see.

- > It is not just for reporting but to find hidden patterns.
- > Also known as knowledge Discovery in Databases.
- > IS used to support decision making.

Example Capital One 2 million transactions and find a hidden pattern: customers withdraw 4 times after a fee increase usually close their account.

Question 15. DATA VISUALIZATION

Data Visualization converts raw data into graphs

- Process of turning numbers into pictures so that is easier to understand.
- It is not the data itself it is the picture of the data.

Example Capital One shows a bar chart where withdrawals in the month of July was the highest, then the manager sees the problem easily.

Question 16 DASHBOARD

- Dashboard is a screen that shows multiple important charts.

- It is designed to organize, visualize, analyze and track data in a centralized manner.

- It is multiple charts together in one place.
Example Branch managers dashboard ~~it~~ shows withdrawals made today with red.

Question 17 Metric

Metric is a single number that is used to measure something.

- Makes raw data clean so that team members can compare results and make choices.
- Metric is not a target, it is a measurement.

Example Total number of withdrawals today is 1500

Question 18 Key Performance Indicator (KPI)

- KPI - is a measure to evaluate company's success.
- used to check if the company is achieving specific goal.

Example Capital KPI

Keep customer withdrawals below R10m per month. if metric is R15m, KPI is falling - red flag.

Question 19 Benchmark

- A process of comparing companies performance to see if you are doing well.

- Company is measured against top competitors.
- It's not your current number

Example Capital compares its withdrawal rate of 15% to industry benchmark for South African banks of 9%. Since 15% is greater than 9% they know they have a problem.

Question 20 Segmentation

- process of dividing customers with similar behaviour into smaller groups so that you can treat the differently.
- Does not treat all customers the same.

Example. customers of caplac are divided into students, low income earners, high earners. They find students withdraw most their money after fee increase, so they create special low-fee account just for students.

Q3 Customer & Sales KPIs

Question 21 Conversion Rate

Measures a percentage of people who completed a desired action.

- calculate how effectively a website turns visitors into people who take specific action.

example. if 500 people visit your online store and 30 buy ~~so~~ something your conversion rate is 6%.

Question 22 Customer Acquisition Cost (CAC)

- CAC is a total cost that the company use to win a new customer. ~~includes~~ advertising and salaries.

example. if a beauty salon in midland spends R10000 on facebook ads and gets 20 new clients. The customer acquisition cost is R500 per clients.

Question 23 Customer Lifetime Value (CLV/LTV)

CLV is the total profit a company can make from one customer over their entire relationship.

- calculate profit the company will make in lifetime from one customer.

Example: Woolworths customers spends R500 per month for 3 years. The CLV is R18000.

Question 24 Churn Rate

- calculate the percentage of customers who stops using a product in a specific period.
- number of customers loss divided by the number of customers in the beginning gives us the Churn Rate.
- is a rate in which a business loses customers over a specific period.

Example: If Netflix had 200 customers and 20 canceled in August.

$$\text{Churn rate} = (200 / 20) \times 100 = 10\%$$

Question 25 Customer Retention Rate

- is the percentage of customers of continued to use a particular product over a specified period.
- indicates customer loyalty to the business.
- Customer Retention rate is the opposite of Churn Rate
- focus on customers staying.

Example: if Churn rate is 10% then retention rate is 90%

Question 26 NET PROMOTER SCORE (NPS)

It's a process of asking customers if they will recommend the company's brand to their loyalty.

-It measures how customers are to recommend the company to others.

Example. After buying at Shein you get an email asking you to rate how likely you are to recommend them.

Question 27 AVERAGE ORDER VALUE (AOV)

AOV tracks the average amount spent whenever a customer makes a purchase.

-It helps business to understand the customers spending behaviour.

-When AOV is high, it means that customers are buying more.

Example. If Temu made R10 000 from 50 orders, last week then the AOV is R200.

Question 28 Sales Funnel

Sales funnel is the steps a customer goes through from the moment they discover a product until they buy it.

Example. A customer sees a caprac global one ad on TV in Midland, visits Midland branch to ask about faces then compares it with standard Bunt, then opens the account.

Q4 MARKETING & Campaign analytics

Question 29. MARKETING Campaign

Marketing campaign is a planned strategy to promote a product to achieve a specific goal.

- promote a specific product across different channels like TV, facebook

Example. In 2023 capitec launched its

live better campaign to promote global one facility for 3 months. the product was advertised on TV, facebook and Instagram videos and in-branah posters in midrand. mall. goal was to get 400 000 new clients.

Question 30. POST Campaign analytics

Post Campaign Analysis is the process of assessing a Market campaign after it ends to calculate what worked, what ~~did not~~ failed and what was achieved.

- Measure Market Campaigning to check where we can improve in the future.

example. After capitec 3 months live better

campaign ended the marketing team ^{did} post campaign analysis. They found. TV ads generated 250000 new accounts with R40 CAC, but facebook generated 15000 with R150 CAC, ~~at~~ In future spend more on TV and fix the app page to reduce bounce rate.

Question 31 Incremental Revenue

Incremental Revenue is the additional income generated because of the campaign

- the extra income a business earns as a result of marketing campaign which is above the normal income it would have made without the campaign.

Incremental Revenue = Revenue during campaign - Revenue without campaign

Example If capital normally makes R 5M per month from new accounts but during the 11th better campaign they made R 10M. The incremental revenue will be R 5M.

Question 32 Return on Investment (ROI)

ROI measures how much profit you make by comparing it with how much you spent.

Formula for calculating ROI = $(\text{Profit} - \text{cost}) / \text{cost} \times 100$

- It calculates profit or loss made by the business

Example . Campaign cost R 10M, profit of R 13M

Then ROI will be $(R 13M - R 10M) / R 10M \times 100 = 30\%$

Question 33 Return on Ads Spend (ROAS)

ROAS - It calculates how much money you earn for every amount you spent on Ads.

- It compares which channels or ads performs better.

- Only focus on Ad income Vs Ad spend

ROAS = Revenue from Ads / Ad spend.

Example . Capital spent R 2M on Google Ads during campaign and those ads generated R 10M in new account revenue.

ROAS = $R 10M / 2M = 5:1$

Question 34. Click Through Rate (CTR)

CTR - is the percentage of people who click on your ads. It shows that people are interested in your product.

formula = clicks / impressions $\times 100$

Example.

capital facebooks ad got 200 000 impressions and 10 000 clicks. $CTR = (10\,000 / 200\,000) \times 100$

$$CTR = 5\%$$

Question 35 Bounce Rate

Bounce Rate - is when people visit the website but does not do anything, they just leave without clicking any ads, or links, they don't take any action.

- People leave the website without visiting other pages.

- High bounce rate shows the page was slow

Example. During the capital live better campaign 65% of visitors left the application page without applying

Question 36 A/B Testing

- check and compare between two ads and tell you which version of ad performed better

- You show different groups of users two versions of something to compare which will perform better.

- A current version and a modified version

Example. Capitalize the better campaign

Version A Red Button Saying Open Account now
at the top of the page
Version B Blue Button Saying Join 10m South
Americans Open Account in 5 minutes at the
button of the page with trust badges.

Question 37 Lead Generation

Lead generation is when the company or business gets the contact details of people who are interested in the business.

- Attract and capture customers contact.

- capture potential customer for future sales.

Example. During the better campaign, Capitalize offered a free money better e-book on facebook in exchange for users to leave their contact numbers. 50000 people filled the form.

Question 38 - Marketing Attribution

- Marketing Attribution is a way to decide which marketing channel deserves to be credited for a sale.

- which channel actually bring more results so that you can invest more there

Example. You see a live better ad on TV, click a google Ad the next day, then watched a facebook review before opening an account. Marketing Attribution helps Capitalize decide how much credit to give TV, google and facebook for that one new account.

References

- Chaffey D and Ellis-Chadwick F 2022. Digital Marketing 6th ed. Harlow: Pearson
- Kotler P, Kartajaya H and Setiawan, I 2021, Marketing 5.0 Technology for Humanity. Hoboken Wiley.
- Laudonk and Taveri C 2023 E-commerce 2023: Business Technology and Society 18th ed. Harlow: Pearson
- Kotler P and Keller K.L. 2020. Marketing management 15th ed. Harlow: Pearson
- Charlesworth A. 2021 Digital marketing. A practical Approach 3rd ed. London Routledge.